PHD125 Pharmaceutical Organic Chemistry · Carbohydrates

After class: check your understanding

Dr. NAM · Faculty of Pharmacy, UiTM Bertam, Penang

About 30 minutes. Questions follow the six subchapters of the lecture and end with a pharmacy scenario.

Answer without notes first, then use the slides to check. The answer key is at the back.

5.1 · Introduction and chirality

Q1 CH5-LO1

Which compound has a chiral carbon?

A. $CH_3CH(OH)COOH$

B. $CH_3CH(OH)CH_3$

C. $HOCH_2COCH_2OH$

D. $HOCH_2CH(OH)CH_2OH$

Q2 CH5-LO1

How many stereoisomers are possible for a molecule with 3 chiral carbons?

Q3 CH5-LO1

Match each term to its meaning.

Options: non-superimposable mirror images · 50:50 mixture of enantiomers · rotates light clockwise (+) · rotates light anticlockwise (−)

enantiomers _____

racemic mixture _____

dextrorotatory _____

levorotatory _____

Q4 CH5-LO1

Esomeprazole and levocetirizine are examples of:

A. single-enantiomer versions of racemic drugs

B. sugar alcohols

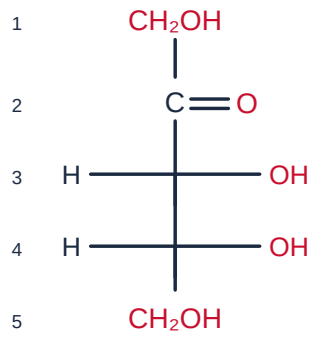
C. disaccharides

D. glycosides

5.2 · Monosaccharides

Q5 CH5-LO2

Classify this sugar.



A. ketopentose

B. aldopentose

C. ketohexose

D. aldotetrose

Q6 CH5-LO2

Is the sugar above D or L?

A. D: the OH on C4 is on the right

B. L: the C=O is at C2

C. D: the C=O is at the top

D. neither: it has no chiral carbon

Q7 CH5-LO2

Match each sugar to its other name.

Options: dextrose (blood sugar) · levulose (fruit sugar) · brain sugar · malt sugar

glucose _____

fructose _____

galactose _____

Q8 CH5-LO2

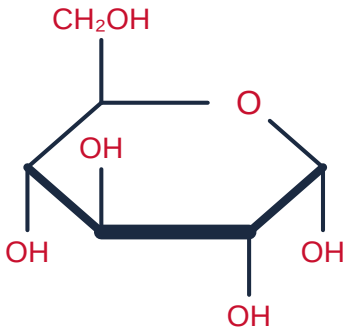
D-glucose and D-galactose differ only in the position of the –OH on C4.

True / False

5.3 · Cyclic forms

Q9 CH5-LO3

Identify this structure.



A. α-D-glucose
B. β-D-glucose
C. β-D-galactose
D. α-D-fructose

Q10 CH5-LO3

In a Haworth projection, a group that was on the right in the Fischer projection points:

A. down
B. up
C. into the ring
D. it disappears

Q11 CH5-LO3

What is the name for the slow change in optical rotation as α- and β-glucose interconvert in water?

_____

Q12 CH5-LO3

Fructose forms a five-membered ring because:

A. the C5–OH adds to the C2 ketone
B. the C6–OH adds to C1
C. it has no carbonyl group
D. it is an L sugar

5.4 · Reactions of monosaccharides

Q13 CH5-LO4

Give the product of each reaction of D-glucose.

Options: gluconic acid · sorbitol · methyl glucoside · fructose

Benedict's reagent (oxidation) _____

H_2/Ni (reduction) _____

methanol, H^+ _____

Q14 CH5-LO4

Fructose gives a positive Benedict's test even though it is a ketose.

True / False

Q15 CH5-LO4

Mannitol given by IV infusion acts as:

A. an osmotic diuretic
B. an anticoagulant
C. a laxative glycoside
D. a sweetener only

5.5 · Disaccharides

Q16 CH5-LO5

Match each disaccharide to its linkage.

Options: α-1,4 · β-1,4 · α-1,β-2 · α-1,6

maltose _____

lactose _____

sucrose _____

Q17 CH5-LO5

Sucrose is a non-reducing sugar because:

A. both anomeric carbons are used in the glycosidic bond

B. it contains fructose

C. it is too sweet

D. it has no –OH groups

Q18 CH5-LO5

What is the name of the 1:1 glucose–fructose mixture made by hydrolysing sucrose?

_____

5.6 · Polysaccharides

Q19 CH5-LO5

Match each polysaccharide to its structure.

Options: unbranched, α-1,4 · branched every 25–30 units · branched every 8–12 units · unbranched, β-1,4

amylose _____

amylopectin _____

glycogen _____

cellulose _____

Q20 CH5-LO5

Iodine gives a red-brown colour with:

A. glycogen
B. amylose
C. cellulose
D. glucose

Q21 CH5-LO5

Which is used as an injectable anticoagulant?

A. heparin
B. dextrin
C. cellulose
D. amylopectin

Pharmacy scenario · Sugar at the counter

Scenario

Encik Hafiz has type 2 diabetes and needs a cough syrup. His daughter, who is lactose intolerant, asks whether the lactose in her antihistamine tablets will upset her. Hafiz also asks why his glucometer strips must only be used for blood glucose.

Q22 CH5-LO5

Which syrup is most suitable for Hafiz?

A. a sugar-free syrup sweetened with sorbitol or an artificial sweetener
B. Syrup BP (about 67% sucrose)
C. a honey-based syrup
D. any syrup: sucrose is not a sugar

Q23 CH5-LO5

Lactose intolerance is caused by:

A. too little of the enzyme lactase
B. a missing enzyme that converts galactose to glucose
C. an allergy to milk protein
D. too much sucrase

Q24 CH5-LO4

Glucometer strips measure only glucose because they use:

A. the enzyme glucose oxidase
B. Benedict's reagent
C. iodine
D. Tollens' reagent

Q25 CH5-LO5

What would you tell Hafiz's daughter about the lactose in her tablets?

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PHD125 Pharmaceutical Organic Chemistry · Carbohydrates

After class: check your understanding

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About 30 minutes. Questions follow the six subchapters of the lecture and end with a pharmacy scenario.

Answer key

5.1 · Introduction and chirality

Q1 CH5-LO1

Which compound has a chiral carbon?

A. $CH_3CH(OH)COOH$

B. $CH_3CH(OH)CH_3$

C. $HOCH_2COCH_2OH$

D. $HOCH_2CH(OH)CH_2OH$

Lactic acid C2: H, OH, CH_3, COOH.

Q2 CH5-LO1

How many stereoisomers are possible for a molecule with 3 chiral carbons?

8

$2^3 = 8$.

Q3 CH5-LO1

Match each term to its meaning.

enantiomers → non-superimposable mirror images; racemic mixture → 50:50 mixture of enantiomers; dextrorotatory → rotates light clockwise (+); levorotatory → rotates light anticlockwise (−)

Ball 16.2.

Q4 CH5-LO1

Esomeprazole and levocetirizine are examples of:

A. single-enantiomer versions of racemic drugs

B. sugar alcohols

C. disaccharides

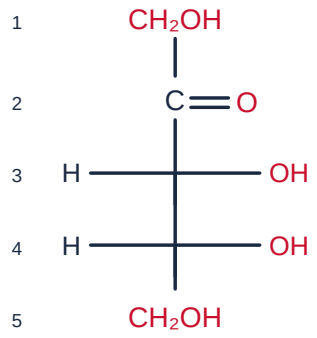
D. glycosides

es- (S) and levo- prefixes mark one enantiomer.

5.2 · Monosaccharides

Q5 CH5-LO2

Classify this sugar.



A. ketopentose

B. aldopentose

C. ketohexose

D. aldotetrose

Five carbons; the C=O is at C2 (ketone).

Q6 CH5-LO2

Is the sugar above D or L?

A. D: the OH on C4 is on the right

B. L: the C=O is at C2

C. D: the C=O is at the top

D. neither: it has no chiral carbon

Look at the chiral carbon furthest from the C=O (C4): OH right → D. It is D-ribulose.

Q7 CH5-LO2

Match each sugar to its other name.

glucose → dextrose (blood sugar); fructose → levulose (fruit sugar); galactose → brain sugar

Malt sugar is maltose (Ball 16.3).

Q8 CH5-LO2

D-glucose and D-galactose differ only in the position of the –OH on C4.

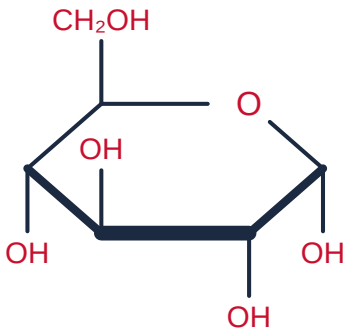
True / False → **True**

They are C4 epimers.

5.3 · Cyclic forms

Q9 CH5-LO3

Identify this structure.



A. α-D-glucose

B. β-D-glucose

C. β-D-galactose

D. α-D-fructose

Six-membered ring, C4–OH down (glucose), C1–OH down (α).

Q10 CH5-LO3

In a Haworth projection, a group that was on the right in the Fischer projection points:

A. down

B. up

C. into the ring

D. it disappears

Right → down, left → up (FGOB 20.3).

Q11 CH5-LO3

What is the name for the slow change in optical rotation as α- and β-glucose interconvert in water?

mutarotation

Equilibrium about 36% α, 64% β.

Q12 CH5-LO3

Fructose forms a five-membered ring because:

A. the C5–OH adds to the C2 ketone

B. the C6–OH adds to C1

C. it has no carbonyl group

D. it is an L sugar

Ring of four carbons and one oxygen: a furanose.

5.4 · Reactions of monosaccharides

Q13 CH5-LO4

Give the product of each reaction of D-glucose.

Benedict's reagent (oxidation) → gluconic acid; H_2/Ni (reduction) → sorbitol; methanol, H^+ → methyl glucoside

Oxidation of –CHO; reduction of C=O; glycoside formation.

Q14 CH5-LO4

Fructose gives a positive Benedict's test even though it is a ketose.

True / False → **True**

In the basic reagent it rearranges to an aldose form (Ball 16.5).

Q15 CH5-LO4

Mannitol given by IV infusion acts as:

A. an osmotic diuretic

B. an anticoagulant

C. a laxative glycoside

D. a sweetener only

It draws water and reduces pressure in the brain.

5.5 · Disaccharides

Q16 CH5-LO5

Match each disaccharide to its linkage.

maltose → α-1,4; lactose → β-1,4; sucrose → α-1,β-2

α-1,6 is the branch link in amylopectin.

Q17 CH5-LO5

Sucrose is a non-reducing sugar because:

A. both anomeric carbons are used in the glycosidic bond

B. it contains fructose

C. it is too sweet

D. it has no –OH groups

Neither ring can open to an aldehyde.

Q18 CH5-LO5

What is the name of the 1:1 glucose–fructose mixture made by hydrolysing sucrose?

invert sugar

It rotates light the opposite way to sucrose.

5.6 · Polysaccharides

Q19 CH5-LO5

Match each polysaccharide to its structure.

amylose → unbranched, α-1,4; amylopectin → branched every 25–30 units; glycogen → branched every 8–12 units; cellulose → unbranched, β-1,4

FGOB 20.7.

Q20 CH5-LO5

Iodine gives a red-brown colour with:

A. glycogen

B. amylose

C. cellulose

D. glucose

Branching breaks up the helix; amylose gives blue-black.

Q21 CH5-LO5

Which is used as an injectable anticoagulant?

A. heparin

B. dextrin

C. cellulose

D. amylopectin

A sulfated polysaccharide.

Pharmacy scenario · Sugar at the counter

Q22 CH5-LO5

Which syrup is most suitable for Hafiz?

A. a sugar-free syrup sweetened with sorbitol or an artificial sweetener

B. Syrup BP (about 67% sucrose)

C. a honey-based syrup

D. any syrup: sucrose is not a sugar

Sucrose and honey raise blood glucose.

Q23 CH5-LO5

Lactose intolerance is caused by:

A. too little of the enzyme lactase

B. a missing enzyme that converts galactose to glucose

C. an allergy to milk protein

D. too much sucrase

The missing galactose enzyme is galactosemia, a different condition.

Q24 CH5-LO4

Glucometer strips measure only glucose because they use:

A. the enzyme glucose oxidase

B. Benedict's reagent

C. iodine

D. Tollens' reagent

Enzymes are specific; Benedict's reacts with any reducing sugar.

Q25 CH5-LO5

What would you tell Hafiz's daughter about the lactose in her tablets?

The amount of lactose in a tablet is usually very small, and most people with lactose intolerance tolerate it without symptoms. If she is very sensitive, a lactose-free brand can be chosen, or she can take the tablet with food. (Galactosemia would be different: that needs strict avoidance.)

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